ASOCIACIÓN ASTRONÓMICA
Y EDUCATIVA DE CANARIAS
Henrietta Swan Leavitt

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E5: Introduction to orbits and black holes

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Worksheet 3: Introduction to black holes

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Solutions

David Sola Gil
(AAEC)

1. The Density of a Black Hole

he Schwarzschild radius for a body of mass M is given by

$$r_s = \frac{2GM}{c^2}.$$

1. Substituting the values of the constants $G = 6.674 \times 10^{-11}\ N \cdot \frac{m^2}{kg^2}$ and $c = 3 \times 10^8\ m/s$ into the given Schwarzschild radius equation, we arrive at

$$r_{s,Sun} \sim 3\ km \ll R_\odot$$
$$r_{s,Earth} \sim 10\ mm \ll R_\oplus$$

As you can observe, in both cases, the Schwarzschild radius is significantly smaller than the actual radius of these celestial bodies. This implies that, when working with the Schwarzschild solution in our daily environment, you should not worry about the event horizon singularity, as it would remain "hidden" inside the object's mass, where the Schwarzschild metric ceases to be valid.

2. For a star with a radius of 10 km and a mass of $1.4M_\odot$, we have

$$r_{s,star} = 1.4 \times r_{s,Sol} \sim 4.2\ km$$

Using the formula for the volume of a sphere of radius r

$$V = \frac{4}{3}\pi r^3$$

The density ρ is given by

$$\rho = \frac{2.8 \times 10^{30}}{\frac{4}{3}\pi {R_\odot}^3} \sim 7 \times 10^{17}\ kg/m^3$$

This density is enormous. To help your students become aware of the magnitude of this figure, I suggest using a striking analogy: this density is equivalent to compressing all of humanity into the volume of a sugar cube. It is an excellent resource for practicing scientific notation and orders of magnitude.

3. To find the mass given the Schwarzschild radius, we solve for M in terms of r_s. We find

$$M = \frac{r_s c^2}{2\,G}$$

Substituting $r_s = 10\ km$ and the remaining constants, we get $M = 3.4\ M_\odot$. If we were to replace the Sun with a black hole of the same mass, the planets in the solar system would continue to orbit the black hole in the same way. This happens because, when gravity is moderate or weak, both Newton and Einstein agree. Extreme effects begin to be observable in more extreme cases, such as near the Schwarzschild radius of a black hole. However, as we have just calculated, the Sun's Schwarzschild radius is "hidden" within its surface.

3. Tidal Forces

As we saw in the notes, both Newton and Einstein agree that the difference in acceleration experienced by two points separated by a distance d, located at a distance r (with $r \gg d$) from the attracting body of mass M is given by

$$\Delta a = \frac{2\,G\,M\,d}{r^3}.$$

1. Assuming a person's height is $d \sim 1.8\,m$, and substituting $M = M_\oplus$, $r = R_\oplus$ and other constants, we have

$$\Delta a_{Human} = \frac{2\,GM_\oplus\,d}{R_\oplus^3} \sim 6 \times 10^{-6}\,m/s^2$$

   Recalling that the acceleration an object experiences on the Earth's surface is $g \sim 10\,m/s^2$, this is completely imperceptible to a human being.

2. In this case, $d = 2R_\oplus$ and the attracting body is the Sun, so $M = M_\odot$ and $r = 150 \times 10^6\,km$. With these values, we find

$$\Delta a_{Earth} = \frac{2\,GM_\oplus\,d}{R_\oplus^3} \sim 10^{-6}\,m/s^2$$

3. For $d = 1.8\,m$, $M = M_\odot$ and $r = 100\,km$, we have

$$\Delta a_{Human} \sim 480\,m/s^2 = 48g.$$

   We would suffer an acceleration difference between our feet and head of about 48 times the acceleration on the Earth's surface. The result is inevitable: immediate spaghettification.

4. For a supermassive black hole, the result changes. For $M = 10^8 M_\odot$, $d = 1.8\,m$ and $r = 295 \times 10^6 + 100\,km$, we have that

$$\Delta a_{Human} \sim 2 \times 10^{-3} m/s^2.$$

   In principle, we could cross the event horizon barely feeling any tidal forces. The problems would come later, as we would inevitably end up plunging toward the singularity. Furthermore, at the moment of crossing the event horizon, we would lose all communication with the outside, since neither light nor any form of communication is able to exit once we have entered.

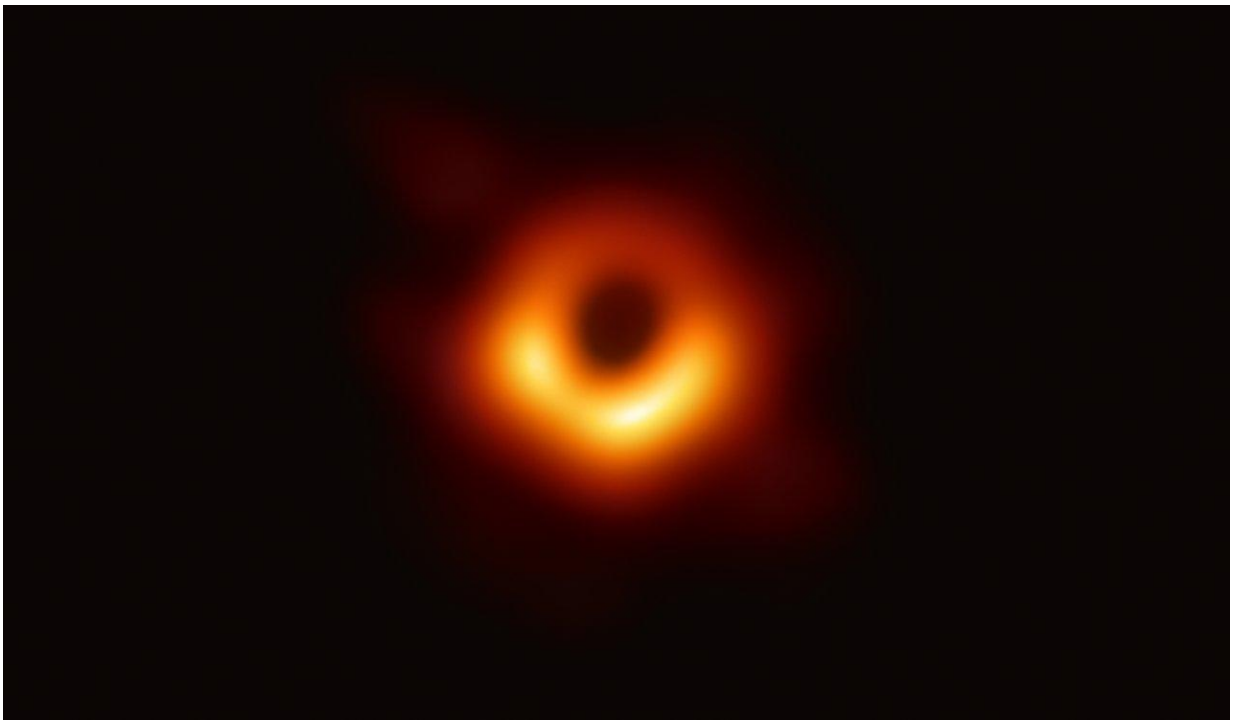
4. Orbits of Light

Null geodesics can be described by the effective potential

$$V_{eff}(r) = \frac{ml^2}{2r^2}\left(1 - \frac{2k}{rc^2}\right).$$

1. For $r \to 0$, the potential diverges towards $-\infty$. For $r \to \infty$, the potential tends to 0. The potential always has negative values, as $r > r_s = 2k/c^2$.

2. The potential always has a maximum at $r = 1.5\, r_s$. This critical point represents *the photon sphere*, the last unstable orbit where light can circulate around the black hole before falling in or escaping.

3. For a body with mass, the point is located at a much greater distance $r = 3\, r_s$, so light is able to get closer to a black hole and return to our eyes.

   What we see in the EHT image of the black hole (below) is not the photon ring, but the *shadow of the black hole*. In a rotating black hole, the shadow is approximately twice the size of the event horizon; we do not see it directly. In practice, light follows curved trajectories, as this GIF illustrates. In practice, light follows curved trajectories, as this GIF illustrates. As a result, much more light than we might initially imagine enters the event horizon and never leaves, leaving a much larger dark zone.